

Stress-testing an income classifier on the Adult Census data

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Dataset Adult Census Income, UCI (roll number ends in 8)
Random seed 138 (last three digits of roll number), 70% train / 30% test
Code <https://github.com/sanskar-94/income-classifier-stress-test>

Every number in this report comes from the notebook `income_classifier_stress_test.ipynb` in the repository above. Running it from top to bottom recreates all tables and figures.

Data preprocessing

The Adult dataset comes from the 1994 US Current Population Survey. Each row is one adult, and the label says whether their yearly income was above \$50K. UCI provides it as two files, `adult.data` (32,561 rows) and `adult.test` (16,281 rows). The assignment asks for my own 70/30 split, so I stacked the two files (48,842 rows) and split them again with `random_state = 138`. Labels in the test file end in a full stop ("`>50K.`"), which I removed so both files use the same two labels.

Missing values

Missing values are coded as "?" and appear in three columns.

Table 1. Missing values in the combined data (48,842 rows).

Column	Missing values	% of rows
workclass	2,799	5.73%
occupation	2,809	5.75%
native_country	857	1.75%

In total 3,620 rows (7.4%) have at least one missing value. I did not delete them, because whether a value is missing is clearly linked to income: only 13.2% of these rows earn `>50K`, compared with 24.8% of complete rows. Deleting them would remove a poorer group and push the base rate up.

Instead I used mode imputation with a missing-indicator flag. Each missing value is filled with the most common category in the training set (Private, Prof-specialty and US), and a 0/1 column records that the value was missing. Occupation is missing in 2,809 rows, and workclass is missing in 2,799 of those same rows. The other 10 are people whose workclass is "Never-worked", so they have no occupation. Because the two columns go missing together, they share one flag, `job_missing`, while native country gets its own flag, `country_missing`. My first attempt gave both columns a separate "Unknown" level, and the logistic regression would not converge. The two Unknown dummies (correlation 0.998) differ only for the 9 training rows whose workclass is "Never-worked", and none of those 9 earns `>50K`, so the logit pushes one coefficient towards plus infinity and the other towards minus infinity. They came out at +14.2 and -15.5 with standard errors of about 5,200, although their sum (-1.30) is stable. That is why I switched to a single flag (the notebook rebuilds this first attempt in Section 1.6).

I also removed 52 rows that were exact duplicates of another row, leaving 48,790.

Columns left out

- `fnlwgt` is the Census sampling weight. It describes how many people a row represents, not the person, so it is not a meaningful predictor.
- `education` is the text form of `education_num` (exactly one label per number), so I kept only the number.
- `relationship` is mostly marital status crossed with sex. "Husband" covers 19,715 men and 1 woman, and "Wife" covers 2,328 women and 3 men. With `sex` and `marital_status` in the model, adding it would make the sex coefficient hard to interpret, because "Husband" and "Wife" already encode sex within marriage. I kept it in the data for one residual check (Question 4) but never used it as a predictor.

Categorical encoding

All categorical columns are one-hot encoded with one level left out as the baseline, so each coefficient compares that level with the baseline. Before encoding I merged categories that were too small to give a stable coefficient:

- `workclass`: Federal, Local and State government became "Government", and "Never-worked" (10) plus "Without-pay" (21) became "No-pay" (31 rows);
- `occupation`: "Armed-Forces" (15 rows) was merged into "Protective-serv";
- `marital_status` went from seven levels to four: Married (including armed-forces spouse), Never-married, Div-Sep (divorced, separated, spouse absent) and Widowed;
- `native_country` went from 41 levels to US / Non-US, since most countries have only a few dozen people;
- `race` kept all five levels.

The baselines are White, Female, Never-married, Private, Prof-specialty and US. For `workclass` and `occupation` the baseline is also the imputed value, so `job_missing` reads as "missing compared with the category it was filled with". The final design matrix has 32 predictors plus an intercept: 5 numeric, 25 dummies and the 2 missing flags.

Outliers

I checked the numeric columns with the $1.5 \times \text{IQR}$ rule but did not delete anything.

- For `hours_per_week` the rule flags 13,486 rows (27.6%), which is everyone outside 32.5 to 52.5 hours. These are part-time workers and people with long weeks, not errors.
- For `age` it flags 215 people aged 79 to 90, and for `education_num` it flags 1,787 people with a level below 4.5 (8th grade or less). Both are plausible values, so I kept them.
- For `capital_gain` and `capital_loss` the IQR is zero, because 91.7% and 95.3% of values are exactly 0. Every non-zero value counts as an "outlier" (4,035 and 2,282 rows), so the rule is no use here.
- `age` = 90 (54 rows), `hours_per_week` = 99 (137 rows) and `capital_gain` = 99,999 (244 rows) are top codes, where the survey caps the value. They are real people with values at or above the cap.

What does need handling is the shape of the two capital columns: a spike at zero and a long right tail. I applied `log1p`, that is $\log(1 + x)$, to both. Zeros stay at zero, and a \$99,999 gain becomes 11.5, so it is no longer 20 times the size of a \$5,000 gain. A few extreme values can then no longer drag the fitted lines around.

Feature scaling and split

The five numeric columns (age, education_num, hours_per_week and the two logged capital columns) are standardised to z-scores. The scaler is fitted on the training rows only and then applied to the test rows, so no information leaks from the test set. The imputation modes also come from the training rows. Dummy variables are left as 0/1. A numeric coefficient is therefore the change for a one standard deviation increase: in log-odds for the logistic model, and in predicted probability for the OLS model. One SD of age is 13.7 years and one SD of education_num is 2.6 levels.

The split gives 34,153 training rows (23.9% earn >50K) and 14,637 test rows (24.0%). All models are fitted on the training rows. Part 1, Part 2.3 and Part 2.4 are evaluated on the test rows. The coefficient comparisons in Parts 2.1 and 2.2 use the training rows, since that is the data the model actually learned from.

Part 1: Build

Model A is ordinary least squares on the 0/1 income label (a linear probability model). Model B is an unregularised logistic regression on the same label. Both use the same 32 predictors and the same training rows. I fitted them with `statsmodels` so I could get standard errors and confidence intervals for later parts. The full coefficient tables are in Appendix A.

Table 2. Model performance. For the logit, the train R^2 row is McFadden's pseudo R^2 .

	Model A (OLS)	Model B (Logit)
Test AUC	0.894	0.898
Test RMSE	0.338	0.327
Train R^2	0.371	0.393 (pseudo)
Test R^2	0.372	n/a
Predictions below 0	21.6%	0.0%
Predictions above 1	0.4%	0.0%
Range of predictions	-0.360 to 1.249	0.000 to 0.997
Accuracy with its own rule	71.3% (median rule)	84.6% ($p \geq 0.5$)

The two models rank people almost identically. Their Spearman rank correlation on the test set is 0.984, and the AUCs are 0.894 and 0.898. Model A has the usual weakness of a linear probability model: its predictions run from -0.360 to 1.249, and 21.6% of them are below zero, so they cannot be read as probabilities.

Comparing the two thresholding rules. For this dataset "high risk" simply means "predicted to earn more than \$50K". The median of Model A's test predictions is 0.2262, and Model A flags everyone at or above it, which by construction is half the test set (7,319 people). Model B flags everyone with $p \geq 0.5$, which is 2,848 people (19.5%).

Table 3. Agreement between the median rule (Model A) and the 0.5 rule (Model B) on the 14,637 test rows.

	B low ($p < 0.5$)	B high ($p \geq 0.5$)	Total
A low ($< \text{median}$)	7,318	0	7,318
A high ($\geq \text{median}$)	4,471	2,848	7,319
Total	11,789	2,848	14,637

The two rules disagree on **4,471 people, 30.55% of the test set**. In every one of these cases Model A says high and Model B says low, and the opposite cell of Table 3 is empty.

Why they differ. The difference comes from where the cut-off sits. The models put people in almost the same order, but 0.2262 on the OLS scale is a much lower bar than 0.5 on the logistic scale, and the median rule flags half the test set when only 24.0% really earn >50K. Since every person Model B flags is also flagged by Model A, the 4,471 mismatches are simply 7,319 minus 2,848. This also explains the accuracy gap in Table 2 (71.3% for the median rule, 84.6% for the 0.5 rule). Question 1 looks at this in more detail.

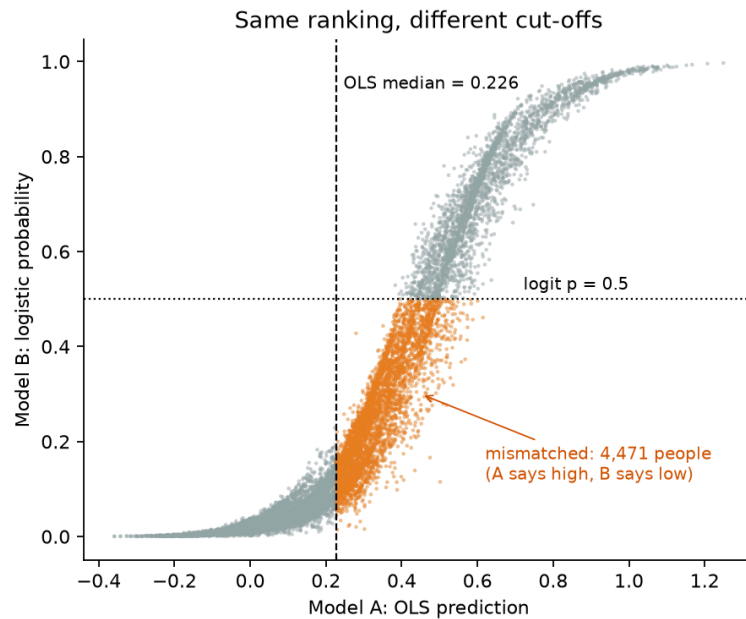


Figure 1. Each dot is a test-set person. The models agree on the ordering (the S-shaped band). The orange points lie between the two cut-offs and are labelled differently by the two rules.

Part 2: The Trap

2.1 Simpson's paradox hunt

I split the training data by marital status into Subgroup A, married (15,735 people), and Subgroup B, unmarried (18,418 people: never married, divorced, separated, spouse absent and widowed). The key predictor is sex (`sex_Male`, with Female as the baseline). For each of the three groups I fitted a crude logit with sex as the only predictor, and an adjusted logit with the full Model B specification. Inside a subgroup, columns that never vary are dropped automatically (for example the marital dummies among married people). In the unmarried subgroup nobody in private household service or the no-pay workclass earns >50K, so their estimates run off to minus infinity and those two dummies were dropped as well.

Table 4. Share earning >50K in the training set, by marital status and sex.

	Women	Men
Married (Subgroup A)	44.7% (n = 1,746)	44.3% (n = 13,989)
Not married (Subgroup B)	4.7% (n = 9,494)	8.3% (n = 8,924)
All training rows	11.0% (n = 11,240)	30.3% (n = 22,913)

Table 5. Logit coefficient on `sex_Male` in the full training set and the two subgroups, with 95% confidence intervals.

Model	n	Crude coef. [95% CI]	Adjusted coef. [95% CI]	p (adjusted)	Odds ratio (adjusted)
Full training set	34,153	+1.262 [1.196, 1.328]	+0.160 [0.064, 0.256]	0.001	1.17
Subgroup A: married	15,735	-0.017 [-0.117, 0.083]	-0.288 [-0.416, -0.161]	< 0.001	0.75
Subgroup B: unmarried	18,418	+0.599 [0.478, 0.720]	+0.740 [0.584, 0.895]	< 0.001	2.10

Result: the trend reverses. In the full training set the sex coefficient is positive, both crude (+1.262) and adjusted (+0.160, CI 0.064 to 0.256). Among married people it turns negative. The adjusted coefficient is -0.288, and its whole confidence interval (-0.416 to -0.161) lies below zero. Among unmarried people it is +0.740 (CI 0.584 to 0.895), stronger than overall. The two subgroup estimates differ by 1.03 ($z = -10.03$, $p \approx 10^{-23}$), and an interaction term fitted in one model gives -1.191 (CI -1.376 to -1.006), so the gap between the subgroups is far too large to be chance.

Two checks give the same answer. Refitting on all 48,790 rows rather than only the training rows gives +0.151 overall, -0.287 for married people and +0.689 for unmarried people. Removing the 160 people in the two categories with no high earners (private household service and no-pay), instead of dropping their columns, moves the unmarried estimate only from 0.740 to 0.729.

Why. Table 4 shows the raw picture. Married people earn >50K far more often than unmarried people, and a much larger share of men than women are married (61.1% vs 15.5%). That composition effect accounts for most of the crude gap. The adjusted reversal has a different cause. Model B fits one sex coefficient for everyone, so its +0.160 behaves like a weighted average of opposite effects in the two groups (weighting -0.288 and +0.740 by their precision gives about +0.13). Strictly, this is effect modification: the overall estimate sits between the two subgroup estimates and one of them runs the other way, which is the reversal the brief asks about. Question 2 works through the numbers. There may also be a selection effect. The extraction kept only people with positive hours worked, so the married women in the data are those in paid work. The near-identical rates for married women and married men (44.7% and 44.3%) are high enough that the label might partly reflect joint income for couples, but the dataset documentation does not settle this.

Policy implication. The adjusted coefficients compare men and women with the same job, hours and education. Those are themselves partly shaped by sex, so the total gap is larger than the adjusted figure suggests (crude odds ratio 3.53). Read on its own, the full model suggests a modest male advantage that applies to everyone (odds ratio 1.17). The subgroup models show an odds ratio of 2.10 among the unmarried and a reversed one (0.75) among the married. On this evidence I would aim measures on the pay gap first at unmarried women (never married, divorced, separated or widowed) rather than at all women equally.

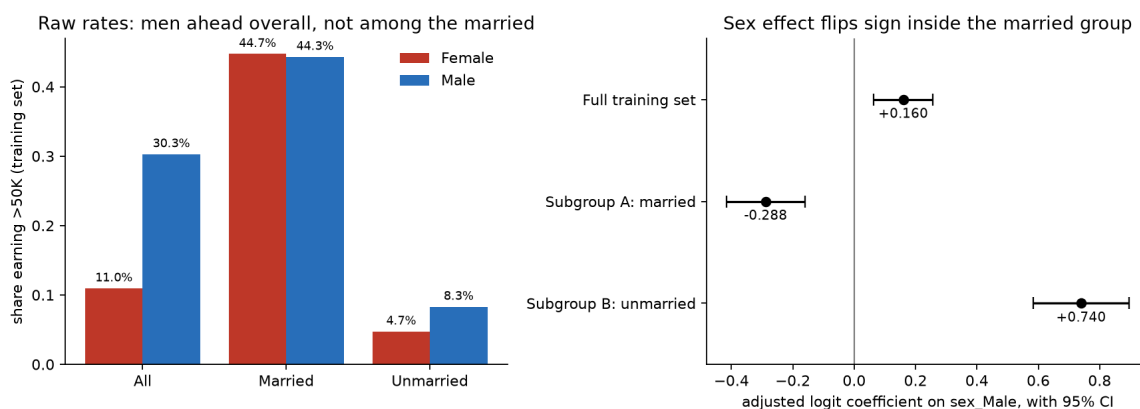


Figure 2. Left: raw >50K rates by sex. Right: adjusted sex_Male coefficient with 95% CI in each group.

2.2 Omitted-variable bias

I dropped marital status (its three dummies). Its married dummy has the largest coefficient of any predictor in Model B (+2.586, an odds ratio of 13.3), and marital status is closely tied to sex, age and race, so I expected that leaving it out would distort the coefficients on those variables.

Table 6. Coefficients for eight predictors before and after removing marital status. % change is $(\text{reduced} - \text{full}) / |\text{full}|$.

Predictor	Full model	Without marital status	Change	% change	Odds ratio
sex_Male	0.160	1.132	+0.972	+607.9%	1.17 → 3.10
age	0.362	0.583	+0.221	+61.2%	1.44 → 1.79
race_Black	-0.237	-0.459	-0.222	-93.7%	0.79 → 0.63
workclass_Self-emp-inc	0.179	0.312	+0.133	+74.0%	1.20 → 1.37
occupation_Exec-managerial	0.212	0.270	+0.059	+27.7%	1.24 → 1.31
hours_per_week	0.368	0.390	+0.021	+5.8%	1.45 → 1.48
education_num	0.757	0.682	-0.075	-10.0%	2.13 → 1.98
capital_gain	0.514	0.498	-0.016	-3.1%	1.67 → 1.65

Which coefficient shifted the most? The coefficient on `sex_Male` shifted most. It rose from 0.160 to 1.132 (+607.9%), so the odds ratio for men went from 1.17 to 3.10. Its absolute change (+0.972) is also the largest of the 29 coefficients the two models share, so the big percentage is not only down to a small starting value. Among the other predictors in Table 6, `race_Black` (-93.7%), incorporated self-employment (+74.0%) and age (+61.2%) moved the most. These big movers all shifted in the direction their link with marriage predicts. Education (-10.0%) and capital gain (-3.1%) shrank slightly instead, even though both are weakly positively correlated with being married. That is a known property of logistic regression: leaving out a strong predictor pulls the remaining coefficients towards zero even when they are unrelated to it (odds ratios are "non-collapsible").

What it reveals. By the usual checks the reduced model is only slightly worse (test AUC 0.852 vs 0.898, accuracy 82.3% vs 84.6%), yet its sex coefficient is now about seven times its full-model value. I would not have noticed this from the performance figures alone.

Reflection. To detect a confound I cannot see, I would run a residual audit: group the residuals by variables the model does not use and look for groups whose average residual is far from zero. Doing this with the dropped `relationship` column showed that wives are under-predicted by 0.084 on average ($t = 5.12$), and a $\text{sex} \times \text{married}$ interaction removed most of that gap. Question 4 sets out the full procedure.

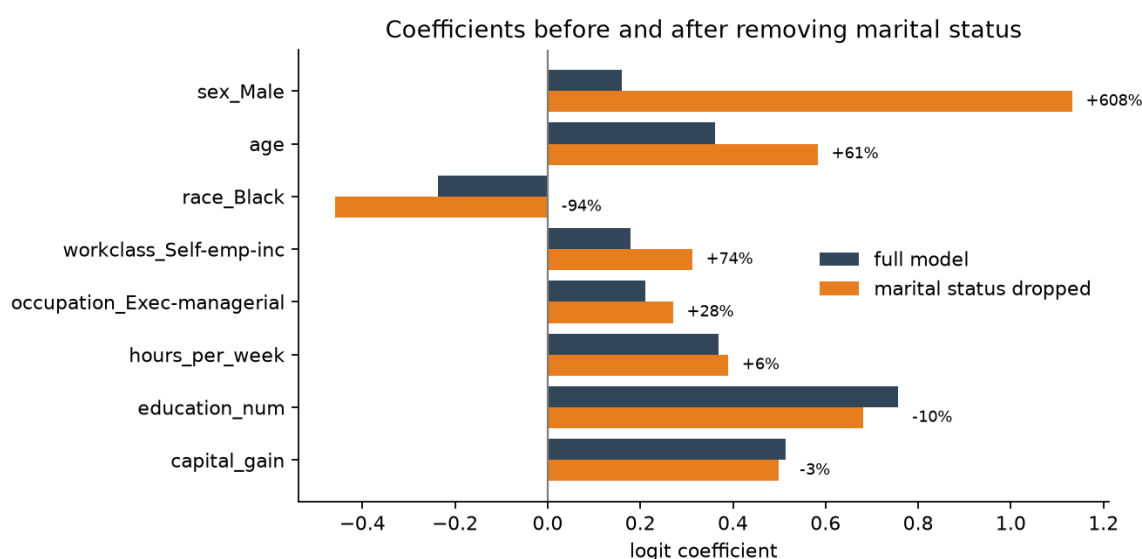


Figure 3. Coefficients in the full model and in the model without marital status.

2.3 Adversarial subset construction

On the test set, Model B makes 215 confident false positives ($p \geq 0.75$ but actually $\leq 50K$) and 623 confident false negatives ($p \leq 0.25$ but actually $> 50K$). That is **838 confident mistakes, 5.73% of the test set**, or 7.70% of the 10,877 predictions the model made with that much confidence. A random sample of 50 (seed 138) contains 37 false negatives and 13 false positives.

Table 7. The 50-row adversarial sample compared with all 838 confident errors and the full test set.

	50-row sample	all 838 confident errors	full test set
n	50	838	14,637
mean age	41.5	43.4	38.8
median age	38.5	41.0	37.0
age under 30 (%)	8.0	11.5	29.5
age 30-49 (%)	64.0	58.6	48.1
age 50+ (%)	28.0	30.0	22.4
male (%)	78.0	75.2	66.3
married (%)	68.0	63.7	45.6
White (%)	90.0	87.5	86.0
Black (%)	4.0	6.9	9.4
born outside US (%)	6.0	9.1	8.4
bachelor's or higher (%)	46.0	35.8	25.2
works 40+ hours (%)	90.0	86.0	75.7
any capital gain (%)	10.0	13.2	8.4
actually earns >50K (%)	74.0	74.3	24.0

Common characteristic. In 49 of the 50 rows (98%) the person's marriage-and-degree profile points the wrong way. Twelve of the 13 false positives are married and have at least a bachelor's degree, a profile held by only 5.3% of all test-set people who earn $\leq 50K$. None of the 37 false negatives has that profile: each is unmarried, has no degree, or both. That half of the trait is a weaker signal, since 59.4% of all high earners in the test set are not married graduates either. Table 7 also shows that the sample is older, more male and more often married than the test set. In addition, 82% are aged 30+ and work 40+ hours (57.3% in the test set), but that mostly mirrors the label mix: 74% of the sample really earn $> 50K$, and 84.3% of all high earners in the test set share it.

Why the model fails confidently. In this additive model the married coefficient (+2.586) is the largest of any predictor; going from high school to a bachelor's degree, for example, adds only about 1.18. A married graduate therefore starts close to the 0.75 line (median p 0.745 in the test set), and a managerial or professional job or long hours can push them over it even when they earn $\leq 50K$. Of the 585 married graduates in the test set who earn $\leq 50K$, 157 end up above 0.75. At the other end, 79.2% of unmarried graduates stay below 0.25. For 23 of the 50 sampled people, marital status was the variable that pushed their prediction furthest in the wrong direction, and occupation came next with 13. Refitting with a $\text{sex} \times \text{married}$ interaction and an age^2 term only cuts the confident errors from 838 to 819. Most of the rest is what the model's own probabilities predict: errors that even a calibrated model makes because the features cannot see everything that drives income. Over the whole $p \leq 0.25$ range the model is calibrated on average (mean prediction 6.5%, actual 6.5%), so about 623 confident false negatives are expected and Model B makes 623, although inside that range the lowest bin is over-predicted and the 0.15 to 0.25 bin under-predicted (Table 8). At the high end the model is overconfident, with 215 confident false positives where about 169 were expected. Question 5 goes through each error type.

Table 8. Calibration of Model B on the test set.

Predicted p	People	Mean predicted p	Actual share >50K
0 to 0.05	5,606	0.018	0.011
0.05 to 0.15	2,550	0.092	0.089
0.15 to 0.25	1,449	0.198	0.231
0.25 to 0.50	2,184	0.360	0.381
0.50 to 0.75	1,576	0.626	0.633
0.75 to 0.90	794	0.819	0.780
0.90 to 1.00	478	0.946	0.916

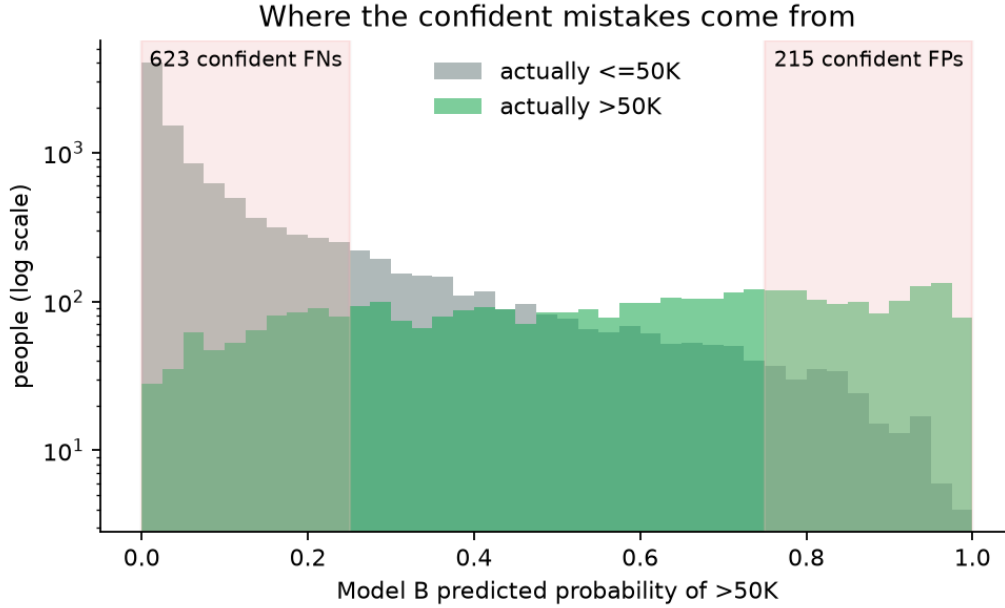


Figure 4. Distribution of predicted probabilities by true class (log scale). The shaded zones contain the confident errors.

2.4 Fairness trade-off analysis

Sensitive variable: sex. Group A is Female (4,936 people in the test set) and Group B is Male (9,701). Their base rates are very different: 10.9% of women and 30.6% of men in the test set earn >50K. I use the three metrics from the brief, each a female/male ratio where 1 is ideal: demographic parity (DP, the ratio of selection rates), equalized odds (EO, the smaller of the true-positive-rate and false-positive-rate ratios) and predictive parity (PP, the ratio of precisions). In the tables TPR, FPR and PPV are the true-positive rate, false-positive rate and precision within each group.

I put women as Group A on purpose. Equalized odds is defined as the *minimum* of two ratios, and with the disadvantaged group in the numerator both ratios are below 1, so the minimum picks the larger gap. With the groups the other way round, the minimum would pick the ratio closer to 1 and hide the problem. To make the metrics concrete I assume the model is used to pre-approve people for a credit product that needs an income above \$50K, so $\hat{Y} = 1$ (being flagged) is the favourable outcome.

Table 9. Fairness metrics across thresholds ($A = \text{Female}$, $B = \text{Male}$). TPR, FPR and PPV are shown per group.

t	DP	EO	PP	Accuracy	TPR F	TPR M	FPR F	FPR M	PPV F	PPV M
0.05	0.600	0.612	0.578	0.614	0.961	0.986	0.363	0.593	0.245	0.424
0.15	0.358	0.320	0.857	0.758	0.809	0.937	0.128	0.401	0.435	0.508
0.25	0.285	0.244	0.929	0.811	0.638	0.856	0.067	0.275	0.538	0.579
0.35	0.260	0.210	0.978	0.837	0.545	0.761	0.039	0.187	0.629	0.643
0.45	0.238	0.189	0.991	0.845	0.441	0.664	0.024	0.128	0.689	0.695
0.55	0.214	0.155	1.024	0.844	0.351	0.570	0.014	0.088	0.759	0.741
0.65	0.187	0.121	1.047	0.835	0.253	0.460	0.007	0.056	0.819	0.783
0.75	0.153	0.098	1.036	0.818	0.147	0.329	0.003	0.030	0.859	0.829
0.85	0.161	0.060	1.069	0.797	0.093	0.192	0.001	0.011	0.943	0.883
0.95	0.113	0.000	1.050	0.774	0.022	0.067	0.000	0.001	1.000	0.952

At $t = 0.95$ only 12 women and 209 men are flagged and no woman is a false positive, so $EO = 0$ and the female PPV of 1.000 rest on very small counts.

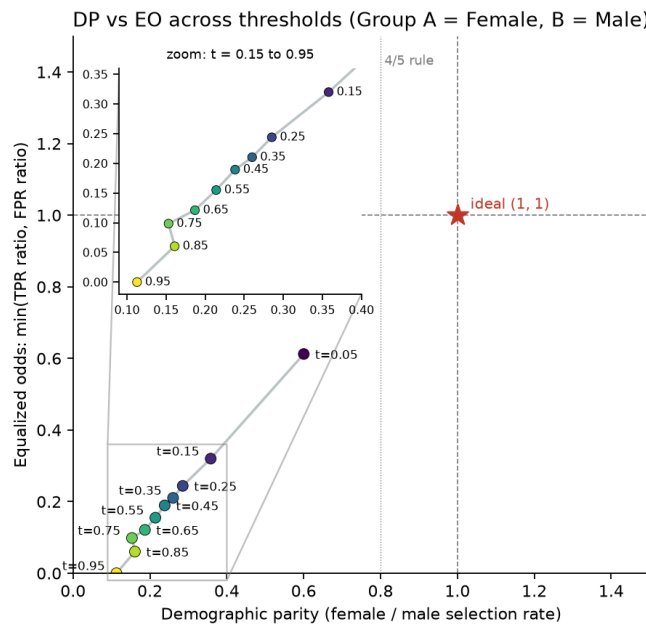


Figure 5. Demographic parity against equalized odds, one point per threshold. The inset zooms into the crowded corner. The ideal is $(1, 1)$.

Quantifying the trade-off. In my data DP and EO mostly move together. Lowering the threshold raises both, because at every threshold the smaller of the two EO ratios is the FPR ratio, and it rises along with the selection rates. There is only one step in the sweep where DP improves while EO gets worse:

If you improve DP from 0.153 to 0.161 (a 5.1% gain, moving from $t = 0.75$ to $t = 0.85$), EO drops from 0.098 to 0.060 (a 38.7% loss). The trade-off cost is $38.7 / 5.1 \approx 7.6$.

That step rests on very few women: at $t = 0.85$ only 53 women are flagged and 3 of them are false positives, so I would not read much into it. In this data the bigger conflict is between DP and EO on one side and PP and accuracy on the other:

If you improve DP from 0.238 to 0.358 (a 50.2% gain, moving from $t = 0.45$ to $t = 0.15$), PP drops from 0.991 to 0.857 (a 13.5% loss). The trade-off cost is $13.5 / 50.2 \approx 0.27$.

On the same move accuracy falls from 84.5% to 75.8%, while EO improves from 0.189 to 0.320.

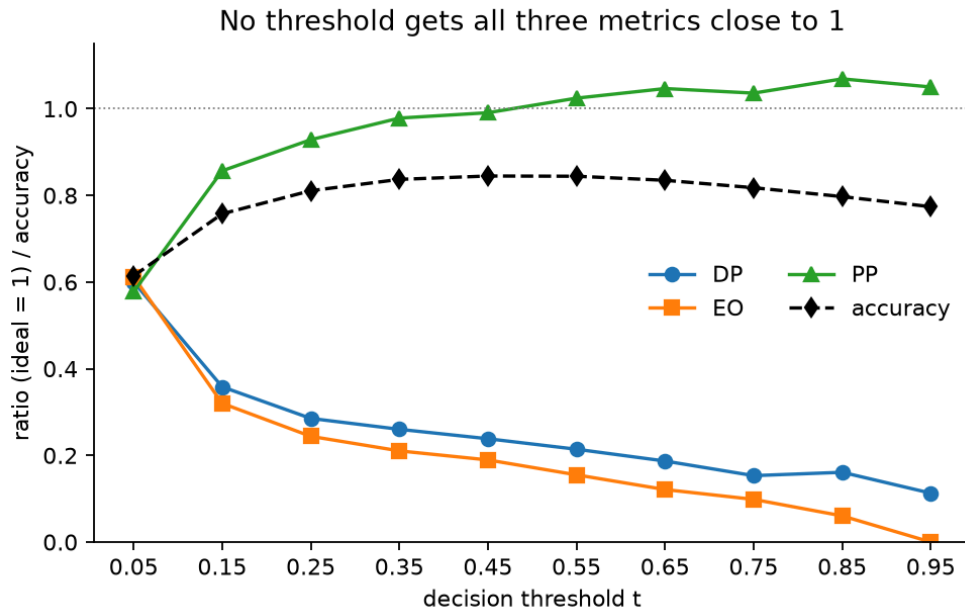


Figure 6. DP, EO, PP and accuracy against the threshold.

Key insight and recommendation. No threshold satisfies all three metrics. DP and EO are best at $t = 0.05$ (0.600 and 0.612), but there PP is 0.578 and accuracy only 61.4%. PP is closest to 1 at $t = 0.45$ (0.991), where DP is 0.238 and EO 0.189. DP never reaches the four-fifths (0.8) line at any threshold. I would recommend $t = 0.35$. It costs little accuracy (83.7%, compared with 84.5% at 0.45, the best in the sweep), keeps PP near parity (0.978) and improves DP and EO over $t = 0.45$. For the policy choice I would prioritise equalized odds over demographic parity, because the base rates are so different (Questions 7 and 8).

Part 3: Descriptive analysis questions

Question 1. Why did the two approaches flag different people?

On the 14,637 test rows the two rules disagree about 4,471 individuals, 30.55% of the test set, and every disagreement is the same kind: high under the OLS median rule and low under the logistic 0.5 rule.

The cause is the cut-off. The median of Model A's predictions is 0.2262, so Model A calls someone high risk once their OLS score reaches 0.2262, while Model B wants a probability of at least 0.5. A median rule flags half the test set by construction (7,319 people), whatever the base rate. Only 24.0% of the test set actually earns more than \$50K, and the 0.5 rule flags 19.5% (2,848). All 2,848 people flagged by Model B are inside Model A's 7,319, so the mismatch is exactly the difference between the two counts.

The models themselves put people in nearly the same order (Spearman correlation 0.984, AUCs 0.894 and 0.898). In this comparison the difference between the models adds no disagreement of its own, because Model A's cut-off is low enough to take in everyone Model B flags. If Model A flagged its top 2,848, the two would still disagree on 420 people, because OLS and the logit weight the predictors slightly differently and so order the borderline cases a little differently. Cutting the OLS scores at 0.5 instead of the median shrinks the mismatch to 483 (3.3%).

The 4,471 people between the two cut-offs are mostly uncertain cases. Their logistic probabilities run from 0.014 to just under 0.5 (median 0.247), and 28.2% of them really earn >50K, only a little above the base rate. The median rule flags all of them because they are in the top half of scores, even though most of them earn ≤50K, and that is why its accuracy is 71.3% compared with 84.6% for the 0.5 rule.

Question 2. The subgroup with a reversed effect

I split by marital status and tracked the coefficient on sex. In the full training set men are far more likely to earn >50K than women (30.3% vs 11.0%), and the crude logit coefficient on sex_Male is +1.262 (95% CI 1.196 to 1.328). Adjusted for the rest of Model B it falls to +0.160 (CI 0.064 to 0.256, $p = 0.001$), still a male advantage.

In the married subgroup (15,735 people) the sign reverses. The adjusted coefficient is -0.288 (CI -0.416 to -0.161), so a married woman has higher odds than a married man with the same profile (odds ratio 0.75). In the unmarried subgroup (18,418 people) the male coefficient is +0.740 (CI 0.584 to 0.895), much stronger than in the full model.

Two things explain this. The crude gap is mostly composition. In the training data 61.1% of men are married compared with 15.5% of women, and married people earn >50K at 44.4% while unmarried people manage 6.5%, so most of the +1.262 is really a marriage gap. The adjusted full model already controls for marital status, though, so composition cannot explain why it gives +0.160 while the married subgroup gives -0.288. That happens because Model B allows only one sex effect for everyone, so +0.160 is a compromise. A model with a sex × married term shows the split directly: +0.801 for unmarried people and $+0.801 - 1.191 = -0.390$ for married people (interaction CI -1.376 to -1.006).

If the subgroups were ignored, a policymaker would see one small male advantage (odds ratio 1.17) and might treat the gap as minor, or aim a single measure at all women. The subgroup models point instead at unmarried women, where men's odds of earning >50K are about twice theirs (odds ratio 2.10), while among married people the adjusted gap runs the other way (possibly because only married women in paid work are in the data).

Question 3. The omitted-variable test

The variable I dropped is marital status: its three dummies (married, divorced/separated, widowed) against never-married. Its married dummy has the biggest coefficient in Model B (+2.586, odds ratio 13.3), and it is tied to several other predictors. A far larger share of men than women are married (61.1% vs 15.5%), married people are older (43.3 vs 34.6 years on average), and Black people are less likely to be married than everyone else (26.9% vs 48.1%).

Coefficient shifts when marital status is removed (same as Table 6).

Predictor	Full model	Without marital status	Change	% change	Odds ratio
sex_Male	0.160	1.132	+0.972	+607.9%	1.17 → 3.10
age	0.362	0.583	+0.221	+61.2%	1.44 → 1.79
race_Black	-0.237	-0.459	-0.222	-93.7%	0.79 → 0.63
workclass_Self-emp-inc	0.179	0.312	+0.133	+74.0%	1.20 → 1.37
occupation_Exec-managerial	0.212	0.270	+0.059	+27.7%	1.24 → 1.31
hours_per_week	0.368	0.390	+0.021	+5.8%	1.45 → 1.48
education_num	0.757	0.682	-0.075	-10.0%	2.13 → 1.98
capital_gain	0.514	0.498	-0.016	-3.1%	1.67 → 1.65

The `sex_Male` coefficient moved the most. It went from 0.160 to 1.132, an increase of 607.9%, which takes the odds ratio for men from 1.17 to 3.10. Its absolute change (+0.972) is also the largest of the 29 shared coefficients. The `race_Black` coefficient became 93.7% more negative, incorporated self-employment rose by 74.0% and age by 61.2%, each in the direction its link with marriage predicts. Education (-10.0%), hours (+5.8%) and capital gain (-3.1%) barely moved.

The reduced model is not simply wrong. Its odds ratio for men (3.10) is close to the raw one (3.53), and it answers a different question: how big the sex gap is when marriage is not held fixed. Strictly speaking, marital status cannot cause a person's sex, so for the sex coefficient it is not a textbook confounder. It is partly a pathway and partly a collider (income also affects who marries), so neither +0.160 nor +1.132 is a clean causal effect. What bothers me is that nothing in the reduced model's performance shows which question it is answering. Its test AUC is 0.852 (full model 0.898) and its accuracy 82.3% (84.6%), so a reader would have no reason to suspect that one coefficient had grown about sevenfold. I would not take any coefficient at face value until I had tried removing and adding likely confounders like this.

Question 4. Detecting a confound I cannot see

I would use a residual audit against variables the model does not contain. If the model misses something important, people who share it will be over- or under-predicted as a group, and their average residual (actual outcome minus predicted probability) will sit away from zero.

I tried it on my own model with `relationship`, the column I dropped during preprocessing. On the test set, people recorded as "Wife" have an average residual of +0.084 (714 people, $t = 5.12$): 46.2% of them earn >50K but the model predicts 37.8%. Husbands sit at -0.003. The model under-predicts wives because it adds the sex and marriage effects separately, the same problem I found with the Simpson's paradox split in Part 2.1. Adding a `sex × married` interaction brings their residual down to +0.006. Own-child ($t = -4.48$), Other-relative ($t = -3.94$) and Unmarried ($t = -2.89$) also stood out before the interaction, and the first two still do after it ($t = -4.60$ and -4.18), so some hidden structure remains.

For a real deployment I would do three things. First, ask domain experts, such as a labour economist or a loan officer, which income drivers the data lacks: industry, weeks worked, region, spouse's income. Second, find

data that has them. The American Community Survey public-use microdata includes industry, weeks worked, region and household income, and the `folktables` package builds Adult-style income tasks from it. It has no capital gain or loss columns, so I would refit without them, using an inflation-adjusted income cut-off, and flag any category whose average residual is far from zero (with a Bonferroni-style threshold, as there are many categories). Third, add each flagged variable and see how far the sex and race coefficients move, as I did with marital status in Part 2.2.

This only finds confounders that someone thought to record, so I would repeat the audit whenever new data becomes available.

Question 5. The 50-row adversarial subset

The 50 rows are a random sample (seed 138) of the 838 confident mistakes in the test set (5.73%): 37 false negatives and 13 false positives.

What they share is a marriage-and-degree profile that points the wrong way, which is true of 49 of the 50 rows (98%). Of the 13 false positives, 12 are married graduates, although only 5.3% of test-set people who earn $\leq 50K$ fit that description. None of the 37 false negatives is a married graduate, which is less telling, as 59.4% of all high earners are not either.

The mechanism is the size of the marriage coefficient in an additive model. Being married adds +2.586 to the log-odds, while going from a high-school diploma to a bachelor's degree adds about +1.18 ($1.56 \text{ SD} \times 0.757$). So the decision boundary is drawn mainly by marriage. A married graduate starts close to 0.75 (median p 0.745), and an executive or professional job or long hours can push them over it; 12 of the 13 sampled false positives have such jobs. Married people without a degree lose much of the boost through education and occupation (machine operators get -0.889; 10 of the 21 married false negatives are machine operators), ending near 0.2. Unmarried people start 2.1 to 2.6 behind, so most stay under 0.25 even with a degree, which 11 of the 16 unmarried false negatives have.

Refitting with a $\text{sex} \times \text{married}$ interaction and an age^2 term only cut the confident errors from 838 to 819, so those two misspecifications explain very little. At the low end the model is calibrated on average: about 623 confident false negatives were expected and 623 occurred. At the top it is overconfident, with 215 false positives where 169 were expected. Most of the rest probably comes from income drivers the data does not contain, such as industry or overtime.

Question 6. What the trade-off means for real people

I took two real people from the test set, matched as closely as I could, in the credit pre-approval setting from Part 2.4.

The woman (39, divorced or separated) really earns more than \$50K and has \$8,614 in capital gains, but the model gives her $p = 0.384$. The man (38, married) has the same education, kind of job, sector and hours but earns less than \$50K, yet gets $p = 0.471$, mostly because of the married coefficient.

At $t = 0.45$, the most accurate threshold in the sweep (84.5%), he is approved and she is rejected, so both decisions are wrong. At that threshold qualified women are approved 44.1% of the time and qualified men 66.4%, and DP is 0.238.

Now suppose a regulator asks for better demographic parity and the bank moves to $t = 0.35$. She is approved, which is correct, and he stays approved, which is still wrong. DP improves from 0.238 to 0.260 and EO from 0.189 to 0.210, but the number of men approved without qualifying rises from 864 to 1,258. Those 394 extra men are given credit they may struggle to repay. On Figure 5 this move is a step up and to the right, so for these two people there is no DP-versus-EO cost; the price is paid in PP and accuracy instead.

Compared with $t = 0.45$, going all the way to $t = 0.05$ raises DP by 151.8% (to 0.600) and lowers PP by 41.7% (to 0.578), a cost of 0.27, the same as in Part 2.4. By then 3,988 men and 1,596 women are approved without qualifying. Only 24.5% of approved women really earn >50K, against 42.4% of approved men, and a lender could use that gap to argue that approving women is riskier. Pushing DP up helped her, but the cost fell on the men approved without qualifying and on approved women as a group.

The two people and their outcome at each threshold ($1 = \text{approved}$).

	Woman (row ID 2713)	Man (row ID 24246)
Age	39	38
Marital status	Div-Sep	Married
Education	Some-college	Some-college
Occupation	Exec-managerial	Exec-managerial
Workclass	Private	Private
Hours per week	40	40
Capital gain	\$8,614	\$0
Actually earns	>50K	≤50K
Model B p	0.384	0.471
Approved at $t = 0.25$	1	1
Approved at $t = 0.35$	1	1
Approved at $t = 0.45$	0	1
Approved at $t = 0.55$	0	0

Question 7. Defending a threshold

I would choose $t = 0.35$, for four reasons.

The accuracy cost is small. At 0.35 accuracy is 83.7%, less than a point below the 84.5% at 0.45 (the sweep's best) and the 84.6% at the default of 0.5.

Predictive parity is close to ideal. PP is 0.978: 62.9% of approved women really earn >50K, against 64.3% of approved men. An approval means nearly the same thing for both sexes, so a lender cannot easily claim that approved women are riskier.

Moving from 0.45 to 0.35 also helps qualified women. The share of them approved goes from 44.1% to 54.5% (TPR ratio 0.664 to 0.716), and DP and EO both improve (0.238 to 0.260 and 0.189 to 0.210). Going lower does not help much: at 0.25 DP only reaches 0.285, while PP drops to 0.929 and accuracy to 81.1%. That is why I stopped at 0.35.

It also suits the setting. The model flags 27.2% of the test set at 0.35, close to the 24.0% who really earn >50K, so it is not wildly over-approving.

The strongest counter-argument is that 0.35 is still very unfair on two of the three measures. A DP of 0.260 means women are approved at about a quarter of the rate of men, far below the four-fifths (0.8) guideline, and a qualified woman is still rejected 45.5% of the time compared with 23.9% for a qualified man. A critic could say I picked the threshold that protects the bank's accuracy and called it fair because one metric looks good. I think that criticism is valid. PP looks good mostly because the model is reasonably well calibrated, and a calibrated model can still treat groups very unequally. Fixing unequal error rates would need a better model or group-specific thresholds, and the latter raise legal problems of their own.

Question 8. Demographic parity or equalized odds?

For an income model used in lending, I would prioritise equalized odds, and in particular its true-positive half (often called equal opportunity): a woman who really earns over \$50K should be as likely to be approved as a man who does.

The base rates differ a lot: 10.9% of women in the test set earn >50K and 30.6% of men. In my sweep DP never goes above 0.600, and it only gets there at $t = 0.05$, where accuracy collapses to 61.4% and PP falls to 0.578. Even a moderate push, from $t = 0.45$ to 0.15, raises DP by 50.2% but costs 13.5% of PP and almost nine points of accuracy (84.5% to 75.8%).

The only step where buying DP costs EO is $t = 0.75$ to 0.85, where DP improves from 0.153 to 0.161 (a 5.1% gain) and EO drops from 0.098 to 0.060 (a 38.7% loss), a cost of 7.6. That loss is entirely in the false-positive half: on the same step the TPR ratio rises from 0.446 to 0.484, and the TPR ratio moves with DP at every step. So for equal opportunity there is no DP trade-off in my sweep; putting EO first mainly means judging the model on its error rates. At $t = 0.35$ EO is only 0.210 and the TPR ratio 0.716. Both are poor, which tells a policymaker that the model itself needs work, while a DP target would only fix the approval rates.

What I am giving up is demographic parity: women would still be approved at about a quarter of the rate of men (DP 0.260). I accept that because income does affect whether someone can repay, and approving people who cannot repay harms them as well as the lender. The 1994 pay gap behind these base rates is partly the result of discrimination, though, so the model should come with policy aimed at that gap.

Part 4: Reflection

"If a policy-maker asked you: 'just give me one number: should we deploy this model or not?', what would you say?"

Direct answer. I would recommend conditional deployment: the model can screen applications, but a person should make the final decision. My headline reason is that its 84.6% accuracy hides the fact that women who really earn over \$50K are approved far less often than men who do.

Why a single number fails. The number a policymaker would most likely ask for is accuracy or AUC. Both look fine for this model, but neither shows the problems I ran into.

Overall accuracy hides who the errors fall on. At $t = 0.5$ the model is more accurate for women than for men, mainly because most women in the data earn under \$50K and the model predicts "low" for nearly all of them. The women it gets wrong are mostly high earners it failed to approve, and in lending that is the mistake I would worry about most.

The threshold is also not something the data can choose. My two models ranked people almost identically, yet the median rule and the 0.5 rule gave different labels to nearly a third of the test set. To pick one, someone has to decide how many wrong approvals are acceptable for each extra qualified woman who gets approved.

A fairness score would not settle it either, because the metrics disagree. Below $t = 0.45$, lowering the threshold improved demographic parity and equalized odds but made predictive parity and accuracy worse, and no threshold in my sweep got all three close to 1. Any combined score would depend on how I chose to weight them.

The confident mistakes are not spread evenly across people. Most of the wrong predictions at $p \geq 0.75$ and $p \leq 0.25$ belong to people whose marital status and education point one way while their income points the other, and at the top end the model is more confident than it should be.

Finally, I cannot rule out confounds that are not in the data. When I removed marital status, a variable I did have, the sex coefficient grew about sevenfold while the model still looked usable. A variable that was never collected could be distorting the model in the same way, and the usual metrics would not warn me.

Grounding in evidence. At $t = 0.5$ accuracy is 91.8% for women and 81.0% for men, but the true-positive rates are 40.1% and 61.8%. The two thresholding rules disagreed on 4,471 people (30.55%) despite a rank correlation of 0.984. Across thresholds DP ranged from 0.113 to 0.600 and PP from 0.578 to 1.069. There were 838 confident errors (5.73% of the test set), and in 49 of the 50 sampled ones marriage and education pointed the wrong way. Dropping marital status moved sex_Male from 0.160 to 1.132 (+607.9%) while AUC only fell from 0.898 to 0.852.

Closing. I expected to finish this assignment with a yes or a no. Most of the answer turned out to depend on choices the data cannot make, mainly the threshold and which fairness definition to use. So rather than one number, I would give the policymaker Table 9 and explain which side of each trade-off I picked, and why.

Appendix A. Model coefficients

Table A1. Coefficients of Model A (OLS) and Model B (logit), fitted on the 34,153 training rows. Numeric predictors are standardised, so their coefficients are per one standard deviation.

Predictor	OLS coef.	Logit coef.	Logit 95% CI	Logit p	Odds ratio
const	+0.1722	-2.817	[-2.960, -2.673]	< 0.001	0.060
age	+0.0296	+0.362	[0.321, 0.402]	< 0.001	1.436
education_num	+0.0798	+0.757	[0.713, 0.801]	< 0.001	2.132
hours_per_week	+0.0331	+0.368	[0.332, 0.405]	< 0.001	1.445
capital_gain	+0.0813	+0.514	[0.484, 0.544]	< 0.001	1.672
capital_loss	+0.0376	+0.235	[0.206, 0.263]	< 0.001	1.265
workclass_Government	-0.0016	-0.019	[-0.118, 0.080]	0.711	0.981
workclass_No-pay	-0.0204	-0.761	[-2.346, 0.824]	0.347	0.467
workclass_Self-emp-inc	+0.0607	+0.179	[0.021, 0.337]	0.026	1.196
workclass_Self-emp-not-inc	-0.0670	-0.513	[-0.632, -0.394]	< 0.001	0.599
marital_status_Div-Sep	+0.0051	+0.450	[0.308, 0.592]	< 0.001	1.569
marital_status_Married	+0.2844	+2.586	[2.472, 2.700]	< 0.001	13.274
marital_status_Widowed	+0.0162	+0.497	[0.224, 0.770]	< 0.001	1.644
occupation_Adm-clerical	-0.1032	-0.507	[-0.650, -0.365]	< 0.001	0.602
occupation_Craft-repair	-0.1270	-0.542	[-0.675, -0.409]	< 0.001	0.582
occupation_Exec-managerial	+0.0288	+0.212	[0.095, 0.328]	< 0.001	1.236
occupation_Farming-fishing	-0.1982	-1.564	[-1.814, -1.313]	< 0.001	0.209
occupation_Handlers-cleaners	-0.1493	-1.249	[-1.500, -0.998]	< 0.001	0.287
occupation_Machine-op-inspct	-0.1511	-0.889	[-1.069, -0.708]	< 0.001	0.411
occupation_Other-service	-0.1184	-1.358	[-1.565, -1.151]	< 0.001	0.257
occupation_Priv-house-serv	-0.0873	-2.576	[-4.153, -0.999]	0.001	0.076
occupation_Protective-serv	-0.0661	-0.190	[-0.408, 0.027]	0.086	0.827
occupation_Sales	-0.0651	-0.341	[-0.472, -0.210]	< 0.001	0.711
occupation_Tech-support	-0.0428	+0.010	[-0.177, 0.197]	0.914	1.010
occupation_Transport-moving	-0.1486	-0.718	[-0.894, -0.543]	< 0.001	0.488
race_Amer-Indian-Eskimo	-0.0467	-0.493	[-0.907, -0.079]	0.020	0.611
race_Asian-Pac-Islander	-0.0108	+0.026	[-0.177, 0.229]	0.802	1.026
race_Black	-0.0194	-0.237	[-0.374, -0.099]	< 0.001	0.789
race_Other	-0.0145	-0.102	[-0.539, 0.335]	0.648	0.903
sex_Male	+0.0370	+0.160	[0.064, 0.256]	0.001	1.173
native_country_Non-US	+0.0005	-0.170	[-0.313, -0.026]	0.020	0.844
job_missing	-0.1374	-1.303	[-1.514, -1.093]	< 0.001	0.272
country_missing	-0.0307	-0.237	[-0.486, 0.012]	0.062	0.789

Appendix B. Reproducing the results

- `income_classifier_stress_test.ipynb` runs the whole analysis from the raw UCI files. Run it with Kernel > Restart & Run All. It downloads the data if the `data/` folder is empty.
- Package versions are pinned in `requirements.txt` (Python 3, pandas, numpy, scikit-learn, statsmodels, scipy, matplotlib).
- Outputs: tables in `outputs/tables/` (CSV), figures in `outputs/figures/` (PNG), and the headline numbers in `outputs/key_numbers.json`.
- The only sources of randomness are the train/test split and the 50-row sample in Part 2.3, and both use `random_state = 138`.